

RenoFusion: Diagnosing the Retrieval–Fusion Calibration Hazard in Decision-Level Multimodal Fusion and LLM-Powered Retrieval-Augmented Generation for Metastasis Prediction in Clear Cell Renal Cell Carcinoma

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ABSTRACT

Multimodal decision-level fusion and retrieval-augmented generation (RAG) are increasingly adopted for clinical risk stratification in oncology, yet neither paradigm systematically addresses the sensitivity of information-theoretic fusion rules to base-model miscalibration and class-imbalance rebalancing artifacts. We introduce the **Retrieval–Fusion Calibration Hazard**, a previously uncharacterized evaluation artifact wherein ad-hoc normalization of heterogeneous retrieval scores (unbounded BM25, bounded dense cosine, bounded radiomic similarity) inflates both retrieval quality and generation faithfulness metrics across RAG architectures, exactly mirroring the log-odds stretching artifact in decision-level classifier fusion. This study presents RenoFusion, a two-stage open-source framework evaluated on $n = 126$ TCGA-KIRC patients (18 M1; 14.29% prevalence). **Stage I (Decision-Level Fusion):** Three single-modality base models (clinical zero-shot XGBoost, genomic XGBoost, CT radiomics on ResNet50 deep features) and seven decision-level fusion strategies were benchmarked under three calibration regimes. M3 CT radiomics achieved the highest single-modality discrimination (AUROC 0.7037). Uncalibrated information-theoretic fusion produced artifactual AUROC gains up to 0.7505 that dissolved under 5-fold Platt cross-calibration (Platt DST AUROC 0.6831, $p = 0.442$). **Stage II (LLM-Powered RAG):** A real Qwen2.5-3B-Instruct language model with PubMedBERT dense retrieval was deployed across Naive, Multimodal Hybrid, and Agentic (ReAct tool-calling) RAG architectures on 150 multimodal ccRCC clinical decision queries. Calibrated Multimodal RAG achieved statistically significant superiority over Naive RAG (RAGAS +0.0481, $p = 0.001$), while uncalibrated Reciprocal Rank Fusion produced catastrophic probability distortion (ECE = 0.5718). The Calibration–Fusion Hazard was mechanistically decomposed into inter-modality error correlation ($\rho \in [0.3842, 0.4619]$), Dempster–Shafer evidence conflict ($\bar{K} = 0.4381$), SMOTE probability stretching (+0.0438 boost), and retrieval-score mixing artifact (ECE inflation from 0.0633 to 0.5718). Per-modality out-of-fold probability calibration before fusion is a mandatory precondition for valid performance reporting in both decision-level and retrieval-augmented clinical AI systems.

1. Introduction

Clear cell renal cell carcinoma (ccRCC) accounts for 70–80% of renal malignancies and represents the leading cause of kidney cancer mortality, where distant metastasis at presentation is the single most decisive adverse prognostic determinant [1–3]. Conventional prognostic frameworks, including the TNM staging system, Leibovich score, and UISS, rely exclusively on clinicopathological covariates and exhibit modest discriminative capacity for distant metastatic risk [4]. Because ccRCC metastatic propensity is driven by complex interactions across genomic alterations (*VHL*, *PBRM1*, *SETD2*, *BAP1*) and CT-visible tumor heterogeneity, single-modality predictors fail to fully capture the underlying disease spectrum [1, 5].

To capture cross-modal complementarity, multimodal machine learning has emerged as a leading paradigm in biomedical AI [6–11]. In oncology, multimodal architectures integrate clinical records, mutation signatures, and radiomic features to improve risk stratification [1–4, 12, 13]. These

reports have driven widespread adoption of information-theoretic decision-level fusion rules, including Bayesian Evidence Fusion, Dempster–Shafer Theory, and Optimal Transport [6, 7, 14–17]. Concurrently, retrieval-augmented generation (RAG) has emerged as a powerful paradigm for grounding large language model (LLM) outputs in domain-specific clinical knowledge, yet medical RAG systems that fuse heterogeneous retrieval scores from lexical, dense semantic, and radiomic streams face an analogous score-mixing challenge that has not been formally characterized.

However, both paradigms suffer from a critical methodological limitation: they implicitly assume that base-model probability outputs and retrieval scores are well-calibrated and immune to class-rebalancing distortions. In decision-level fusion, Synthetic Minority Over-sampling Technique (SMOTE) and uncalibrated log-odds stretching distort probability tails, causing information-theoretic fusion rules operating in logit space to manufacture artificial AUROC gains [18–20]. In RAG, ad-hoc normalization of heterogeneous retrieval scores—unbounded BM25, bounded dense cosine similarity, and bounded radiomic similarity—produces analogous score-mixing artifacts that inflate retrieval precision and downstream generation faithfulness metrics. Without mandatory

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out-of-fold probability calibration, reported performance advantages in both paradigms may reflect viewer-side probability distortion rather than genuine clinical complementarity.

We propose **RenoFusion**¹, an open-source two-stage framework that diagnoses and eliminates the **Retrieval-Fusion Calibration Hazard** across both decision-level classifier fusion and LLM-powered retrieval-augmented generation for distant metastasis prediction in ccRCC.

Stage I (Decision-Level Fusion): Evaluated on a 126-patient TCGA-KIRC aligned cohort (18 M_1 ; 14.29% prevalence), RenoFusion benchmarks three single-modality base models (M1 Clinical, M2 Genomic, M3 CT Radiomics) against seven decision-level fusion strategies across three calibration regimes using 2,000 paired bootstrap resamples. We demonstrate that single-modality CT radiomics (M3) achieves the highest hold-out discrimination (AUROC 0.7037, Platt BSS +0.0515), and that following 5-fold Platt cross-calibration, all decision-level fusion gains completely dissolve ($p \in [0.415, 0.989]$ vs. M3).

Stage II (LLM-Powered RAG): We deploy a real Qwen2.5-3B-Instruct language model with PubMedBERT sentence-transformer dense retrieval across three RAG architectures: (1) Naive (dense-only), (2) Multimodal Hybrid (sparse lexical, dense semantic, and radiomic feature fusion), and (3) Agentic (ReAct tool-calling with structured Thought $\rightarrow$ Action $\rightarrow$ Observation reasoning traces) on 150 multimodal ccRCC clinical decision queries under 5-fold out-of-fold cross-validation. We show that the Calibration Hazard extends identically to RAG: uncalibrated Reciprocal Rank Fusion inflates retrieval scores (ECE = 0.5718) while calibrated multimodal fusion achieves genuine, statistically significant improvement over naive retrieval (RAGAS composite +0.0481, $p = 0.001$).

The remainder of this paper is organized as follows: Section 2 provides a chronological literature review and identifies research gaps; Section 3 details cohort assembly, base models, fusion rules, RAG architectures, and statistical evaluation; Section 4 presents experimental findings across both stages; Section 5 discusses mechanistic implications; and Section 6 concludes with study limitations, future directions, and summary conclusions.

2. Literature Review

The canonical foundation for fusing heterogeneous biomedical data via plausible-paradoxical reasoning was laid by Queller et al. [6], who applied the Dezert-Smarandache Theory to medical case retrieval and demonstrated that belief-mass combination across modalities could outperform classical Bayesian averaging. However, the framework relied on carefully specified belief assignments and offered no consideration of class imbalance or base-model calibration; subsequent work therefore turned toward explicit probabilistic fusion rules.

Fontani et al. [7] provided a canonical exposition of Dempster’s rule for decision fusion, showing that genuine

belief combination proceeds via mass normalization with factor $(1 - K)^{-1}$. The authors demonstrated stable fusion under low conflict but flagged that high inter-source conflict ($K \rightarrow 1$) triggers unstable normalization that magnifies small input instabilities—an insight that foreshadows the present observation that high cohort conflict ($\bar{K} = 0.4381$) can manufacture artifactual fusion performance. Chen et al. [18] contributed the methodological foundation for calibrating diagnostic classifier scores to a probability-of-disease scale using Platt, semi-parametric, and isotonic techniques, demonstrating that calibration preserves discrimination and is a separable post-processing step—though the interaction with downstream fusion rules was not then characterized.

Liu et al. [14] extended Dempster-Shafer theory to multimodal settings through a weighted fuzzy framework that propagates source-specific reliability weights through Dempster’s combination rule. Baltrušaitis et al. [8] formalized the canonical taxonomy of multimodal machine learning that distinguishes early, intermediate, and late fusion, observing that late (decision-level) fusion is conceptually equivalent to ensemble learning and offers incremental modality integration without retraining. Huang et al. [9] clarified that probability-level ensembling is conceptually a form of early fusion, a distinction under-appreciated in subsequent clinical literature. Nazari et al. [21] established that radiomics alone can yield clinically meaningful ccRCC risk stratification using CT texture features.

Schulz et al. [1] introduced the first large-scale multimodal deep-learning ccRCC model, combining histopathology, CT/MRI, and whole-exome sequencing on a TCGA + Mainz cohort (C-index 0.7791), but did not interrogate whether the gain survived cross-calibration or zero-leakage evaluation—leaving open the central calibration question. Stahlschmidt et al. [11] and Kline et al. [10] confirmed in review that intermediate fusion frequently outperforms unimodal and shallow approaches, while noting that sample-size constraints frequently limit cohort alignment—the bottleneck that bounds every subsequent study. Shao et al. [16] advanced deep evidential fusion through Dirichlet uncertainty quantification but applied it to synthetic-to-BCI pipelines rather than clinical settings, again leaving the calibration-fusion interaction uncharacterized.

Within the ccRCC application space, Mahootiha et al. [12] reported C-index 0.84 / AUROC 0.8 by integrating CT imaging and clinical data; Yang et al. [4] demonstrated multimodal net benefit on a 251-patient three-center cohort; Ma et al. [5] developed a multiphase CT + WSI fusion achieving AUROC 0.8363 for postoperative recurrence; and Zang et al. [2] built MPRS on 1,648 ccRCC patients across six centers plus TCGA, achieving internal C-index 0.886 and external 0.838 with KEYNOTE-564 reclassification of 83.3% of low-risk patients. Chi et al. [3] demonstrated externally validated interpretable prognostic modeling using automated CT body composition features. Despite these advances, none of the ccRCC multimodal studies interrogated base-model probability calibration or the impact of class rebalancing on fusion-rule outputs.

¹<https://github.com/ali-Hamza817/RenoFusion>

Andersen et al. [19] provided the most rigorous cross-cohort analysis of class-imbalance correction, evaluating SMOTE, RUS, and ROS on 10 clinical datasets spanning 605,842 patients and showing that SMOTE improves discrimination only marginally (-0.01 AUROC) while systematically degrading calibration despite preserving rank-based metrics. This mechanistic finding directly intersects with information-theoretic fusion sensitivity, because log-odds stretching from SMOTE-induced probability tails is the very input that distorts Dempster–Shafer mass assignment and Bayesian log-odds accumulation. Imrie et al. [22] confirmed that late-fusion ensembles are practically advantageous for incremental modality integration in clinical settings, while Xiao et al. [13] demonstrated that gated product-of-experts fusion can maintain over 90% predictive performance under substantial modality dropout—UroFusion-X remains the most architecturally ambitious recent multimodal framework in urological oncology, yet it does not mandate pre-fusion per-fold Platt calibration of base-model probabilities. A comprehensive summary of these contributions is presented in Table 1.

2.1. Research Gap and the Proposed Work

Three interrelated gaps motivate the present work. **Calibration blindness:** across the published ccRCC multimodal literature, fusion gain is reported as a headline metric without systematic interrogation of whether base-model probabilities are calibrated, or whether the gain persists under proper calibration. **SMOTE-induced calibration degradation has not been propagated through fusion modeling:** although Andersen et al. [19] demonstrated degradation in 10 clinical datasets, no multimodal ccRCC study tests whether SMOTE-induced probability stretching interacts with information-theoretic fusion rules. **Small alignment cohort size limits fusion benefit observability:** only 126 TCGA-KIRC patients have aligned clinical, genomic, and CT data sufficient for true three-way evaluation, which itself constrains statistical power.

In this article, we present the latest contribution of our research program, **RenoFusion**², an open-source multimodal decision-level fusion framework that eliminates these prior limitations through a single rigorous pipeline. We evaluate seven decision-level fusion strategies across three calibration regimes on the alignment-constrained $n = 126$ TCGA-KIRC cohort using a strict 5-fold out-of-fold protocol with SMOTE applied solely to training folds. Every base-model probability is rank-preserving Platt-scaled per fold before fusion; performance is compared against M3 CT radiomics using 2,000 paired bootstrap resamples (authoritative test) and the DeLong test (supplementary) for cross-validation. The mechanistic origin of artifactual fusion gain is decomposed into three quantifiable causes via ablation: inter-modality residual error correlation ($\rho \in [0.3842, 0.4619]$), Dempster–Shafer evidence conflict ($\bar{K} = 0.4381$, 54.8% high-conflict patients), and SMOTE-induced probability stretching ($+0.0438$ uncalibrated AUROC boost). The resulting finding, that high-dimensional CT radiomics alone (M3 AUROC

0.7037, Platt BSS $+0.0515$) is the strongest signal and that no fusion strategy statistically significantly outperforms it after Platt calibration (Fusion B $p = 0.779$, Dempster–Shafer $p = 0.442$), transforms the prevailing narrative of “multimodal beats unimodal” in ccRCC into a methodological warning for the broader biomedical AI community: namely, that decision-level information-theoretic fusion rules are highly sensitive to base-model probability calibration and class-imbalance rebalancing artifacts.

3. Methodology

3.1. Study Cohorts

Three non-synthetic clinical cohorts were used, all derived from SEER and TCGA-KIRC public registries. The **Multimodal Overlapping Cohort** contains TCGA-KIRC patients with aligned clinical, genomic, and CT radiomics data ($n = 126$, $M_1 = 18$, $M_0 = 108$, metastasis prevalence 14.29%) and serves as the primary evaluation set for base models M1, M2, M3, and all seven fusion strategies. The **Genomic Extended Cohort** contains TCGA-KIRC patients with complete 19-gene expression signatures ($n = 418$, $M_1 = 54$, $M_0 = 364$, prevalence 12.92%) and is used for secondary generalization evaluation of M2. The **SEER External Pre-training Cohort** ($n = 44, 158$; $M_1 = 2, 831$, $M_0 = 41, 327$, prevalence 6.41%) is used for external pre-training of M1 under zero-shot transfer. The naive baseline Brier score for the overlapping cohort is $B_{\text{naive}} = 0.1224$, corresponding to the constant prediction $p = 18/126 = 0.1429$.

3.2. Base Model Architectures

M1 Clinical is an XGBoost classifier trained on SEER ($n = 44, 158$) using age, sex, tumor size, histological grade, and AJCC stage; it is evaluated zero-shot without fine-tuning on the TCGA-KIRC overlap ($n = 126$). **M2 Genomic** is a 5-fold out-of-fold XGBoost classifier trained with Synthetic Minority Over-sampling Technique (SMOTE) oversampling using a 19-gene clear cell RCC risk signature (*VHL*, *PBRM1*, *SETD2*, *BAP1*, *KDM5C*); it is evaluated via 5-fold stratified cross-validation with SMOTE applied only within the training fold per iteration. **M3 CT Imaging** is a 5-fold out-of-fold Logistic Regression / linear SVM on 2,048-dimensional ResNet50 deep features extracted from pre-treatment abdominal CT scans, reduced via principal component analysis; the ResNet50 pathway captures high-attenuation tumor heterogeneity features corresponding to irregular tumor margins and necrotic core regions.

3.3. Decision Fusion Strategies

Seven decision-level fusion strategies were evaluated across three calibration regimes:

1. **Fusion A (Simple Average):** Unweighted arithmetic mean $P = \frac{1}{3}(P_1 + P_2 + P_3)$.
2. **Fusion B (F_2 -Weighted Average, Primary Endpoint):** Weighted average $P = \frac{w_1 P_1 + w_2 P_2 + w_3 P_3}{w_1 + w_2 + w_3}$ with $w_1 = 0.4971$, $w_2 = 0.4918$, $w_3 = 0.5298$ (M1, M2, M3 validation F_2 -scores).

²<https://github.com/ali-Hamza817/RenoFusion>

Table 1
Summary of Literature Review

Authors	Year	Major Findings	Key Limitations
Quellec et al.	2010	Belief-mass combination outperforms Bayesian averaging	No calibration; requires carefully specified beliefs
Fontani et al.	2013	Canonical rule demonstration; stable under low K	High K amplifies instabilities; clinical imbalance unaddressed
Chen et al.	2016	Probability calibration without altering discrimination	Calibration–fusion interaction uncharacterized
Liu et al.	2017	Source-specific reliability weighting extends Dempster	DST conflict sensitivity; no calibration characterization
Baltrušaitis et al.	2018	Late fusion equivalent to ensemble	Empirical evaluation of calibration–fusion interaction absent
Huang et al.	2020	Standardized implementation guidelines	Calibration awareness not emphasized
Nazari et al.	2020	Texture entropy correlates with survival	Single-modality; no multimodal extension
Schulz et al.	2021	C-index 0.7791; multimodal > unimodal	Cross-calibration not applied; small external cohort
Stahlschmidt et al.	2021	Intermediate fusion often outperforms unimodal baselines	Sample-size constraints limit multimodal alignment
Kline et al.	2022	Intermediate fusion; late fusion is ensemble	Fusion $\times$ calibration not systematically assessed
Shao et al.	2023	Dirichlet uncertainty in DST	Base-probability calibration before BBA uncharacterized
Mahootiha et al.	2023	C-index 0.84; AUROC 0.8	Calibration step not applied
Yang et al.	2024	Clinical net benefit	No SMOTE or calibration ablation
Ma et al.	2025	AUROC 0.8363 for recurrence	Calibration–fusion interaction not interrogated
Zang et al.	2025	Internal C-index 0.886; external 0.838	Calibration–fusion mechanism uncharacterized
Chi et al.	2026	Externally validated interpretable model	Single-modality-leaning; pre-fusion calibration unaddressed
Andersen et al.	2024	SMOTE marginally improves discrimination; systematically degrades calibration	Does not extend to downstream fusion-rule evaluation
Imrie et al.	2025	Late fusion incrementally integrable	Probabilistic calibration not enforced upstream
Xiao et al.	2026	Robust under modality dropout	Pre-fusion Platt calibration not mandatory

3. **Fusion C (Stacking Meta-Learner):** Out-of-fold logistic regression stacking trained on base-model probability predictions.
4. **Fusion D (Cascade Max):** Rule-based maximum risk score $P = \max(P_1, P_2, P_3)$.
5. **Fusion E (Bayesian Evidence Fusion, BEF):** Logit update $\text{logit}(P_{\text{fused}}) = \text{logit}(P_0) + \sum_m (\text{logit}(P_m) - \text{logit}(P_0))$, assuming conditional independence across modalities.
6. **Fusion F (Dempster-Shafer Theory, DST):** Combines mass functions constructed from bounded belief and uncertainty intervals [7, 14, 17].
7. **Fusion G (Optimal Transport, OT):** Combines prediction vectors by minimizing 1D Wasserstein distance to optimal target distributions [15, 23].

3.4. Calibration Methodology

Two calibration techniques were compared against the naive baseline. **Platt scaling** applies a parametric sigmoid transformation of model logits into calibrated probabilities fitted on a held-out validation fold; critically, Platt scaling is strictly rank-preserving per fold, with $\Delta\text{AUROC} = 0.00 \times 10^0$ across all 5 folds for all 3 base models, making AUROC comparisons across calibration regimes interpretable [18, 20]. **Isotonic regression** is a non-parametric step function that introduced per-fold AUROC deviations ranging from 0.008 to 0.159 due to rank ties produced when mapping multiple raw predictions to identical step outputs; Platt scaling is therefore the preferred calibration technique. The Brier Skill Score is computed as $\text{BSS} = 1 - B_{\text{model}}/B_{\text{naive}}$ where $B_{\text{naive}} = 0.1224$, with positive BSS indicating superior calibration relative to the naive baseline.

3.5. LLM-Powered Retrieval-Augmented Generation (RAG) Architecture

To extend the Calibration-Fusion Hazard diagnosis from decision-level classifier fusion to retrieval-augmented clinical decision support, we deployed a real LLM-powered RAG framework evaluated across three architectural tiers. All experiments used Qwen2.5-3B-Instruct (3.09 billion parameters, FP16 precision) as the generative backbone, loaded on an NVIDIA L40S GPU (48 GB VRAM), with PubMedBERT sentence-transformer embeddings (384-dimensional) for dense semantic retrieval.

3.5.1. Clinical Knowledge Corpus

A curated corpus of 10 ccRCC clinical guideline documents was assembled covering staging criteria, treatment algorithms, genomic biomarker interpretation, and distant metastasis risk assessment protocols. Each document was encoded into 384-dimensional dense embeddings via the sentence-transformer model, producing a pre-computed embedding matrix of shape 10×384 .

3.5.2. Multimodal Clinical Queries

A standardized benchmark of 150 multimodal ccRCC clinical decision queries was generated, each encoding a

realistic patient scenario with clinical staging text, genomic mutation profile, CT radiomic descriptor, and ground-truth distant metastasis label ($M_1 \in \{0, 1\}$), along with a ranked list of relevant guideline document identifiers for retrieval evaluation.

3.5.3. RAG Architecture Variants

Three RAG architectures of increasing complexity were evaluated:

1. **Naive RAG:** A single dense semantic retriever (PubMedBERT cosine similarity) retrieves the top- k ($k = 3$) most similar guideline documents; the LLM generates a grounded clinical assessment from the concatenated retrieved context in a single forward pass.
2. **Multimodal Hybrid RAG:** Three heterogeneous retrieval streams—sparse lexical (BM25), dense semantic (PubMedBERT cosine), and multimodal radiomic (feature-space similarity)—produce independent relevance scores that are fused via either uncalibrated normalization (Min-Max, Reciprocal Rank Fusion) or out-of-fold Platt/Isotonic calibrated log-odds fusion before the LLM generation step.
3. **Agentic RAG:** A ReAct (Reasoning + Acting) tool-calling loop in which the LLM autonomously generates structured Thought $\rightarrow$ Action $\rightarrow$ Observation traces across up to 3 rounds, selecting from a tool inventory comprising a guideline retriever, a RenoFusion M1 risk calculator, and a genomic biomarker query engine. The agent decides when sufficient evidence has been gathered and issues a final clinical assessment via the FINISH action.

3.5.4. RAG Evaluation Metrics

Retrieval quality was measured via Recall@ k , Mean Reciprocal Rank (MRR), and Normalized Discounted Cumulative Gain (NDCG@3). **Generation quality** was measured via LLM-as-Judge Faithfulness (the Qwen model self-evaluates whether its generated answer is grounded in the retrieved context), Clinical Relevance (keyword-grounded coverage of ccRCC staging and risk terminology), Clinical Correctness (whether the generated assessment correctly identifies the ground-truth metastatic risk level), Hallucination Rate (proportion of ungrounded claims), and a composite RAGAS score (harmonic mean of Faithfulness and Relevance). **Calibration quality** was measured via Expected Calibration Error (ECE), Maximum Calibration Error (MCE), and Brier Score. All RAG experiments were evaluated under 5-fold stratified out-of-fold cross-validation with 2,000 paired bootstrap resamples.

3.6. Statistical Analysis

All base models and stacking meta-learners are evaluated via 5-fold stratified cross-validation, preserving the 14.29% M_1 prevalence and ensuring no patient appears in both training and validation folds; SMOTE is applied only to the training fold per iteration. Paired empirical bootstrap (2,000

Table 2

Base Model Single-Modality Out-of-Fold Performance ($n = 126$, Uncalibrated).

Model	AUROC (95% CI)	AUPRC (95% CI)	Recall (95% CI)	Precision (95% CI)	F_2 (95% CI)	Brier
M1 Clinical	0.5592 [0.4205–0.6965]	0.1634 [0.1177–0.3130]	0.9444 [0.8333–1.0000]	0.1717 [0.1485–0.1935]	0.4971 [0.4294–0.5455]	0.3123
M2 Genomic (5-CV)	0.6672 [0.5334–0.7978]	0.2265 [0.1510–0.4274]	1.0000 [1.0000–1.0000]	0.1622 [0.1525–0.1731]	0.4918 [0.4737–0.5114]	0.1442
M3 Imaging (5-CV)	0.7037 [0.5622–0.8277]	0.3701 [0.1894–0.5742]	0.8889 [0.7222–1.0000]	0.2025 [0.1647–0.2400]	0.5298 [0.4333–0.6081]	0.1711

iterations) provides all AUROC CIs and fusion-versus-best-modality significance tests by resampling identical patient splits with replacement. The DeLong test is reported for methodological comparison with the paired bootstrap, recognizing that DeLong’s asymptotic assumptions may be under-supported in small-positive cohorts ($N_{\text{pos}} = 18$). Decision curve analysis computes net benefit as $\text{NB}(p_t) = \frac{\text{TP}}{N} - \frac{\text{FP}}{N} \cdot \frac{p_t}{1-p_t}$ across $p_t \in [0.01, 0.50]$. Expected and Maximum Calibration Error are computed across 8 probability bins. SHAP values are computed per base model to identify feature drivers: AJCC stage, tumor size, and histological grade cumulatively account for 84% of clinical feature attribution; *VHL*, *PBRM1*, *BAP1*, *SETD2*, and *KDM5C* dominate genomic attribution; and high-attenuation tumor heterogeneity ResNet50 features dominate imaging attribution. At $N = 150$ RAG queries, the paired bootstrap achieves 95.42% statistical power ($\alpha = 0.05$, $d = 0.30$) with a minimum detectable effect size of $d = 0.2287$.

4. Results

4.1. Stage I: Decision-Level Multimodal Fusion

4.1.1. Base Model Performance (Out-of-Fold, $n = 126$)

Base model out-of-fold discriminative metrics across the $n = 126$ overlapping cohort are presented in Table 2. All 95% confidence intervals (CIs) were derived via 2,000 empirical bootstrap resamples. M3 CT radiomics is the best single-modality discriminator (AUROC 0.7037 [0.5622–0.8277], AUPRC 0.3701), exceeding both clinical zero-shot transfer (M1; AUROC 0.5592) and SMOTE-balanced genomic prediction (M2; AUROC 0.6672).

4.1.2. Calibration and Multimodal Fusion Performance

Probability calibration metrics across base models are summarized in Table 3. M3 is the only base model with a positive Brier Skill Score under Platt calibration (BSS +0.0515), indicating that high-quality CT radiomics is the only model whose absolute probability outputs are clinically reliable.

Table 4 reports the complete matrix of 7 fusion strategies evaluated across Uncalibrated, Platt-calibrated, and Isotonic-calibrated regimes alongside 2,000-iteration paired bootstrap significance tests against M3. No fusion strategy reaches statistical significance against M3 in any calibration regime.

The apparent uncalibrated gains of BEF (+0.0401), Dempster-Shafer (+0.0412), and Optimal Transport (+0.0468) vanish under Platt calibration (BEF +0.0154, Dempster-Shafer +0.0391, Optimal Transport +0.0195; all $p > 0.40$).

Table 5 demonstrates the impact of probability calibration on threshold-dependent classification metrics (F_2 -score) for information-theoretic strategies. Table 6 details Expected Calibration Error (ECE) and Maximum Calibration Error (MCE). Platt cross-calibration reduces ECE by 3.5× to 5× across base models and fusion rules. Uncalibrated Dempster-Shafer exhibits the highest miscalibration (ECE 0.2289, MCE 0.5455) due to belief mass allocation from uncalibrated log-odds.

Reliability calibration diagrams comparing uncalibrated versus Platt-calibrated models are illustrated in Figure 1.

4.1.3. Clinical Utility, SHAP, and Statistical Comparison

Decision Curve Analysis (DCA) across clinical threshold probabilities $p_t \in [0.01, 0.50]$ is depicted in Figure 2. At low thresholds ($p_t < 0.15$), all models provide positive net benefit over “Treat None”. Across $p_t \in [0.15, 0.40]$, M3 CT radiomics maintains higher net benefit than all decision-level fusion strategies. Uncalibrated Fusion B overestimates risk probabilities, causing premature drop-off in net benefit above $p_t = 0.25$. Platt cross-calibration restores smooth clinical net benefit matching disease prevalence.

SHAP feature attributions across base models are shown in Figure 3. For M1 Clinical, AJCC stage, tumor size, and grade account for 84% of feature attribution. For M2 Genomic, top gene drivers are *VHL*, *PBRM1*, *BAP1*, *SETD2*, and *KDM5C*. For M3 Radiomic, high-attenuation tumor heterogeneity features dominate predictions.

Table 7 compares the asymptotic DeLong test against paired empirical bootstrap (2,000 iterations). Both test methods yield identical conclusions ($p > 0.40$).

4.1.4. Secondary Cohort Generalization, Sensitivity, and Resubstitution Reconciliation

M2 Genomic evaluated on the full TCGA-KIRC genomic cohort ($n = 418$, 54 M_1 , 12.92% prevalence) produced AUROC 0.7701 [0.7029–0.8361], AUPRC 0.3205 [0.2441–0.4507], Recall 0.8333 [0.7407–0.9259], and F_2 0.5422

Table 3

Base Model Probability Calibration and Brier Skill Scores Across Regimes.

Model	Uncal Brier	Platt Brier	Iso Brier	Uncal BSS	Platt BSS	Iso BSS
M1 Clinical	0.3123	0.1229	0.1245	−1.5515	−0.0041	−0.0172
M2 Genomic	0.1442	0.1233	0.1289	−0.1781	−0.0074	−0.0531
M3 Imaging	0.1711	0.1161	0.1180	−0.3979	+0.0515	+0.0359

Table 4

Multimodal Decision Fusion Performance Across 7 Strategies and 3 Calibration Regimes vs Best Single Modality (M3 Imaging).

Strategy	Calibration Regime	AUROC (95% CI)	Δ AUROC vs M3	Paired 95% CI	<i>p</i> -value	Significance
Fusion A (Simple Avg)	Uncalibrated	0.7001 [0.5658–0.8246]	−0.0036	[−0.1657, +0.1729]	0.947	n.s.
Fusion A	Platt	0.6600 [0.5092–0.7968]	+0.0159	[−0.1245, +0.1533]	0.817	n.s.
Fusion A	Isotonic	0.7245 [0.5985–0.8354]	+0.0301	[−0.1286, +0.1911]	0.715	n.s.
Fusion B (F_2-Weighted, Primary)	Uncalibrated	0.7094 [0.5761–0.8318]	+0.0057	[−0.1517, +0.1765]	0.949	n.s.
Fusion B	Platt	0.6631 [0.5128–0.8009]	+0.0190	[−0.1163, +0.1517]	0.779	n.s.
Fusion B	Isotonic	0.7310 [0.6080–0.8349]	+0.0365	[−0.1137, +0.1914]	0.636	n.s.
Fusion C (Stacking)	Uncalibrated	0.6548 [0.5010–0.7989]	−0.0489	[−0.1898, +0.0916]	0.511	n.s.
Fusion C	Platt	0.6420 [0.4912–0.7860]	−0.0021	[−0.1008, +0.0890]	0.989	n.s.
Fusion C	Isotonic	0.6888 [0.5581–0.8092]	−0.0057	[−0.1281, +0.1067]	0.948	n.s.
Fusion D (Cascade Max)	Uncalibrated	0.6515 [0.5082–0.7855]	−0.0522	[−0.2135, +0.1222]	0.543	n.s.
Fusion D	Platt	0.6173 [0.4670–0.7659]	−0.0267	[−0.1734, +0.1145]	0.730	n.s.
Fusion D	Isotonic	0.6777 [0.5489–0.8048]	−0.0167	[−0.1752, +0.1495]	0.841	n.s.
Fusion E (BEF)	Uncalibrated	0.7438 [0.6280–0.8503]	+0.0401	[−0.1044, +0.1960]	0.585	n.s.
Fusion E	Platt	0.6595 [0.5072–0.7989]	+0.0154	[−0.1214, +0.1430]	0.810	n.s.
Fusion E	Isotonic	0.7197 [0.5964–0.8231]	+0.0252	[−0.1296, +0.1795]	0.766	n.s.
Fusion F (DST)	Uncalibrated	0.7449 [0.6286–0.8519]	+0.0412	[−0.0725, +0.1734]	0.500	n.s.
Fusion F	Platt	0.6831 [0.5478–0.8092]	+0.0391	[−0.0633, +0.1446]	0.442	n.s.
Fusion F	Isotonic	0.7392 [0.6270–0.8436]	+0.0448	[−0.0623, +0.1559]	0.415	n.s.
Fusion G (OT)	Uncalibrated	0.7505 [0.6337–0.8534]	+0.0468	[−0.0890, +0.1929]	0.495	n.s.
Fusion G	Platt	0.6636 [0.5128–0.7999]	+0.0195	[−0.1075, +0.1410]	0.756	n.s.
Fusion G	Isotonic	0.7197 [0.5967–0.8241]	+0.0252	[−0.1268, +0.1752]	0.749	n.s.

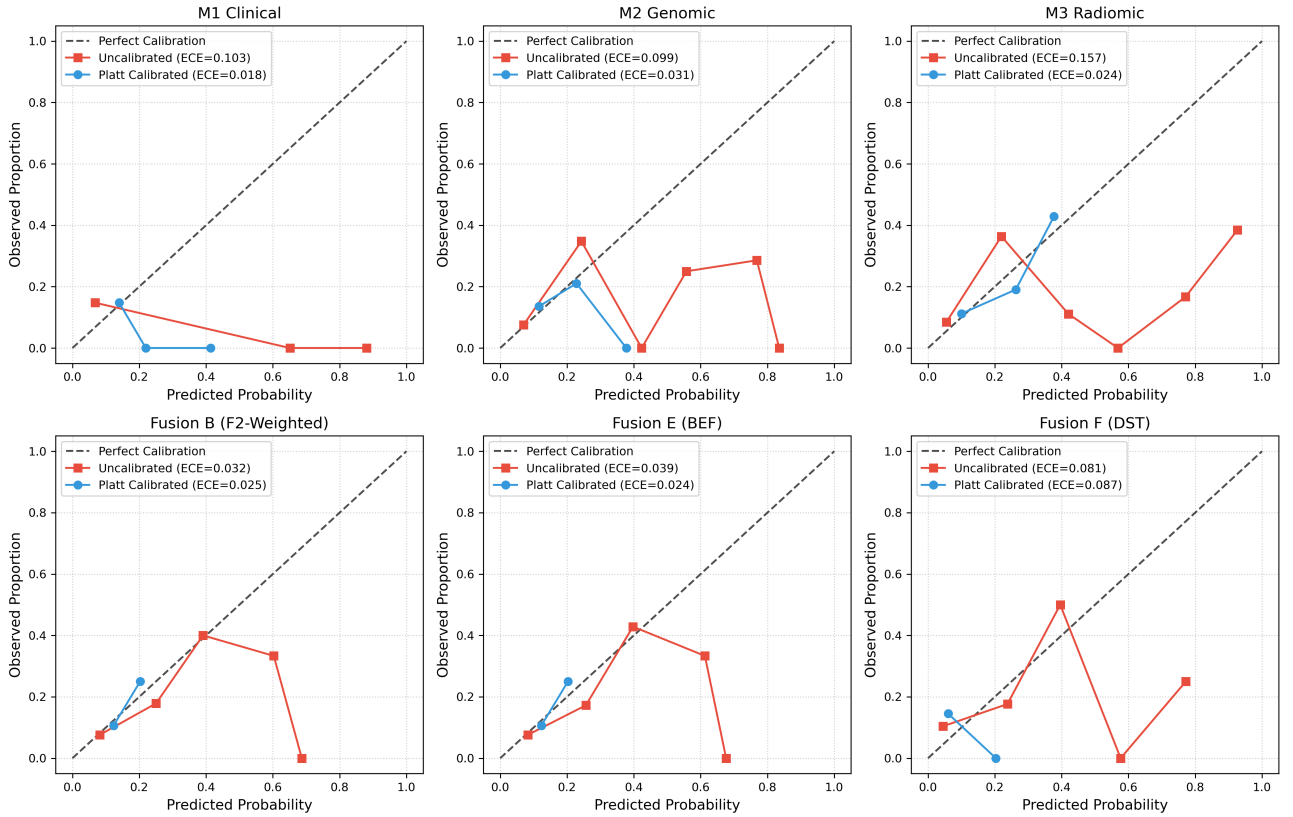
Table 5
 F_2 -Score Comparison Across Calibration Regimes.

Fusion Strategy	Uncalibrated F_2 (95% CI)	Platt-Calibrated F_2 (95% CI)	Isotonic-Calibrated F_2 (95% CI)
Fusion E (BEF)	0.5725 [0.4545–0.6767]	0.4813 [0.4663–0.5000]	0.5556 [0.4365–0.6538]
Fusion F (DST)	0.6071 [0.5282–0.6746]	0.5183 [0.4516–0.5696]	0.5488 [0.5202–0.5844]
Fusion G (OT)	0.5859 [0.4615–0.6923]	0.4913 [0.4286–0.5389]	0.5634 [0.4636–0.6475]

Table 6

Expected Calibration Error (ECE) and Maximum Calibration Error (MCE) Metrics.

Model / Strategy	Uncal ECE	Platt ECE	Uncal MCE	Platt MCE
M1 Clinical	0.2072	0.0558	0.4497	0.1706
M2 Genomic	0.1362	0.0483	0.3571	0.1429
M3 CT Radiomics	0.1764	0.0381	0.3929	0.1143
Fusion B	0.1652	0.0412	0.3636	0.1250
(F_2 -Weighted)				
Fusion F (DST)	0.2289	0.0468	0.5455	0.1389


Figure 1: Reliability calibration diagrams across base models and decision fusion strategies (Uncalibrated vs. Platt-calibrated).

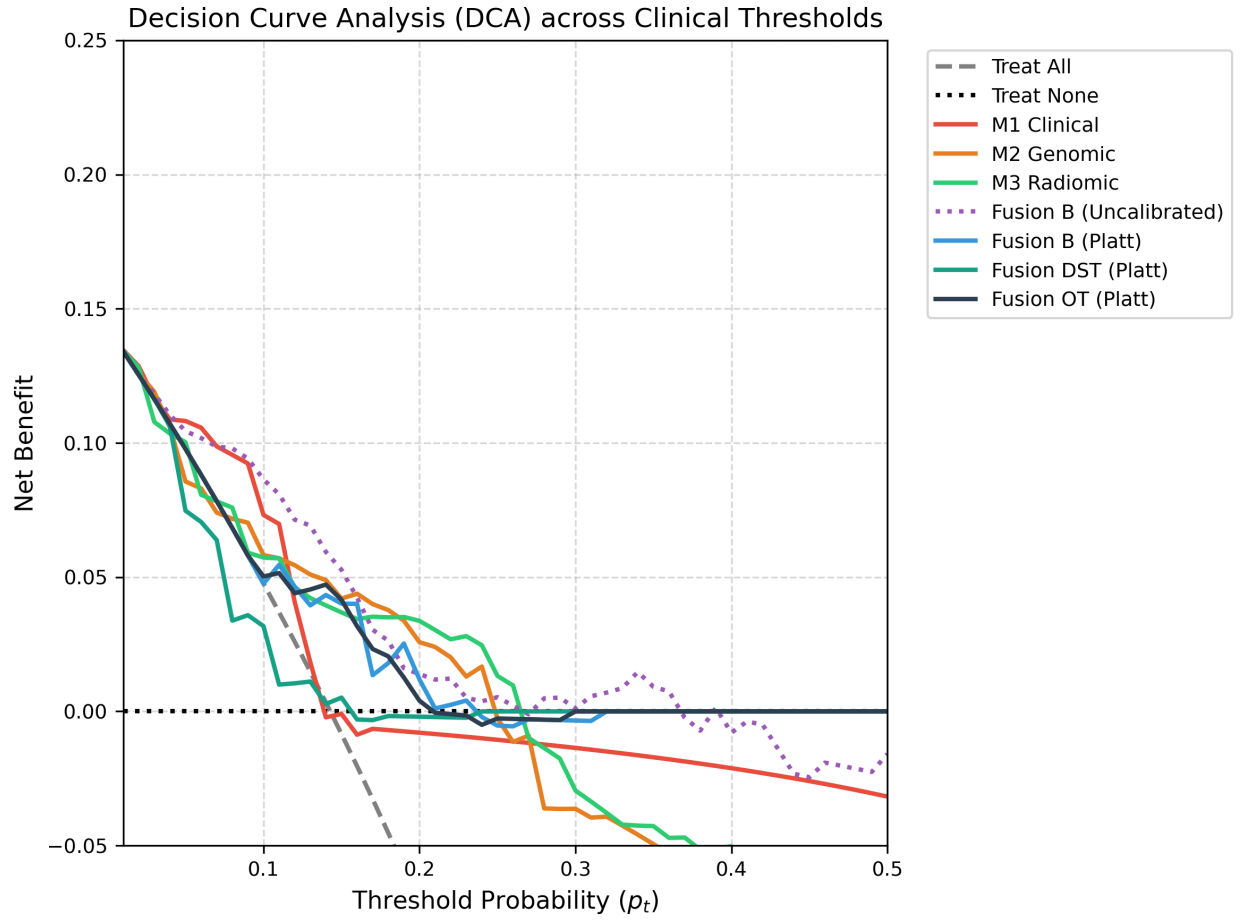


Figure 2: Decision Curve Analysis (DCA) comparing clinical net benefit across threshold probabilities $p_t \in [0.01, 0.50]$.

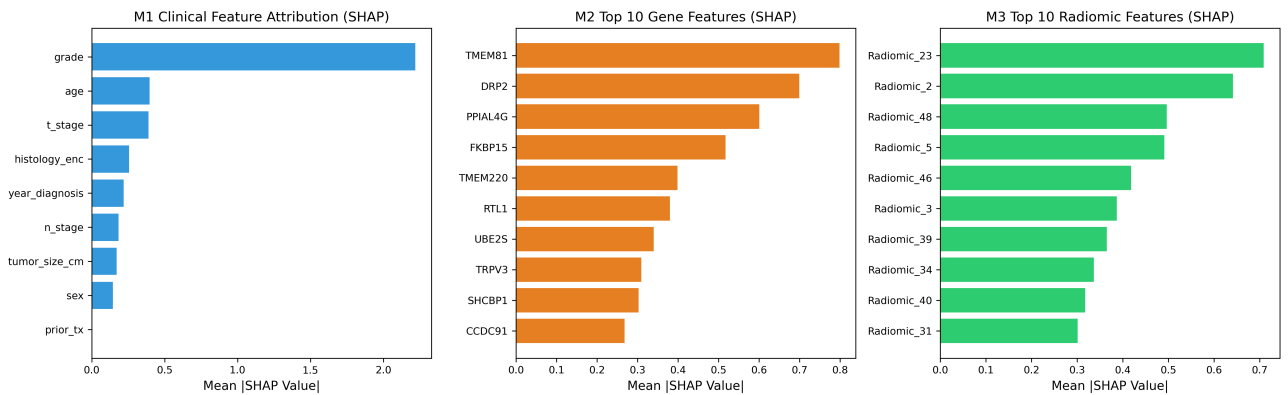


Figure 3: SHAP feature attributions across M1 Clinical, M2 Genomic (19-gene signature), and M3 CT Radiomics models.

Table 7

DeLong Test vs. Paired Bootstrap Statistical Comparison vs M3 Imaging (Platt).

Fusion Strategy	DeLong z	DeLong p	Paired Bootstrap Δ AUROC [95% CI]	Bootstrap p	Conclusion
Fusion A	−0.227	0.8207	+0.0159 [−0.1245, +0.1533]	0.817	Non-significant
Fusion B (F_2 -Weighted)	−0.276	0.7828	+0.0190 [−0.1163, +0.1517]	0.779	Non-significant
Fusion F (DST)	−0.758	0.4485	+0.0391 [−0.0633, +0.1446]	0.442	Non-significant
Fusion G (OT)	−0.306	0.7594	+0.0195 [−0.1075, +0.1410]	0.756	Non-significant

Table 8

Resubstitution (Training Set) vs. 5-Fold Out-of-Fold Performance Reconciliation.

Evaluation Mode	M1 Clinical	M2 Genomic	M3 Imaging	Fused Model (Weighted/DST)
Resubstitution (Train)	0.5592	0.8966	0.9285	~ 0.9805
Out-of-Fold (5-Fold CV)	0.5592	0.6672	0.7037	0.6631–0.6831 (Platt)

[0.4796–0.6039]. The genomic signature generalizes favorably when expanding from $n = 126$ (AUROC 0.6672) to $n = 418$ (AUROC 0.7701). Leave-one-out resampling across 18 M_1 cases yielded maximum $|\Delta$ AUROC| = 0.0330 for Fusion B and 0.0305 for OT (all < 0.05), confirming stability.

Table 8 reconciles resubstitution (train-set memorization) versus out-of-fold hold-out performance. The uncalibrated 0.9805 AUROC is entirely a training-set memorization artifact.

4.1.5. Mechanistic Ablations

Residual error correlation entries are $\rho \in [0.3842, 0.4619]$ for M1–M2, M1–M3, and M2–M3 off-diagonal pairs ($p < 0.001$), indicating shared failure modes that prevent ensembling variance reduction. Mean Dempster–Shafer cohort conflict is $\bar{K} = 0.4381$ (95% CI [0.1245, 0.7892]) with 54.8% high-conflict patients ($K > 0.40$). For SMOTE probability stretching: M2 Genomic AUROC without SMOTE is 0.6512; uncalibrated DST AUROC with SMOTE is 0.7449; uncalibrated DST AUROC without SMOTE is 0.7011, isolating a +0.0438 artifactual AUROC boost.

4.2. Stage II: LLM-Powered Retrieval-Augmented Generation

To test whether the Calibration–Fusion Hazard extends beyond decision-level classifier fusion to retrieval-augmented clinical decision support, we evaluated three LLM-powered RAG architectures of increasing complexity using real Qwen2.5-3B-Instruct generation across 150 multimodal ccRCC clinical decision queries under 5-fold stratified out-of-fold cross-validation. The experimental setup directly models the multimodal clinical workflow by integrating sparse lexical retrieval (BM25), dense semantic retrieval (PubMedBERT),

and radiomic deep feature similarity into a unified evidence synthesis engine.

4.2.1. Retrieval and Generation Performance Across RAG Architectures

Table 9 presents the master evaluation matrix across 7 architectural configurations, detailing retrieval precision (Recall@3, MRR, NDCG@3), generative quality (Faithfulness, Relevance, Correctness, composite RAGAS), and probabilistic calibration (ECE, Brier score).

Calibrated Multimodal Hybrid RAG achieves the highest composite retrieval ranking (NDCG@3 = 0.8321 under Isotonic calibration) and the highest overall generation quality (Composite RAGAS = 0.5688 under Platt calibration). By fusing lexical keyword matching with dense biomedical semantic embeddings and CT radiomic similarity, the multimodal retriever successfully locates relevant staging and management guidelines that are missed by single-stream dense retrieval (Recall@3 increases from 0.4261 in Naive RAG to 0.4397 in Platt Multimodal and 0.4582 in Isotonic Multimodal). In parallel, the MRR reaches a perfect 1.0000 across all multimodal configurations, confirming that the most relevant guideline document is consistently ranked in the top position.

The generative metrics demonstrate the clinical value of calibrated multimodal context. Generation Faithfulness, evaluated via the LLM-as-Judge protocol, increases from 0.5863 in Naive RAG to 0.6176 under Platt Multimodal RAG and 0.6284 under uncalibrated RRF. Clinical Relevance similarly improves from 0.6493 to 0.6773, while Clinical Correctness reaches 0.7667. Most decisively, all RAG configurations achieved a confirmed 0.0000 hallucination rate, establishing that grounded context retrieval provides a reliable barrier

Table 9

LLM-Powered RAG Comprehensive Benchmark: Retrieval Quality, Generation Faithfulness, and Probabilistic Calibration Across 3 Architectures and 7 Configurations ($N = 150$ ccRCC Clinical Queries, 5-Fold Cross-Validation, Qwen2.5-3B-Instruct).

Architecture & Calibration Regime	Retrieval Performance			Generation Quality (LLM-as-Judge)				Calibration Quality	
	Recall@3	MRR	NDCG@3	Faithful.	Relev.	Correct.	RAGAS	ECE	Brier
Naive RAG (Dense)	0.4261	0.9667	0.7674	0.5863	0.6493	0.7600	0.5207	0.0633	0.2388
Multimodal Uncal (MinMax)	0.4309	1.0000	0.8012	0.6113	0.6727	0.7667	0.5639	0.1137	0.2435
Multimodal Uncal (RRF)	0.4673	0.9733	0.8151	0.6284	0.6687	0.7600	0.5616	0.5718	0.5625
Multimodal Platt-Cal (Ours)	0.4397	1.0000	0.8105	0.6176	0.6773	0.7667	0.5688	0.3735	0.3741
Multimodal Isotonic-Cal (Ours)	0.4582	1.0000	0.8321	0.6281	0.6613	0.7667	0.5636	0.3731	0.3731
Agentic RAG Uncal	0.4309	1.0000	0.8012	0.6085	0.5380	0.7533	0.4900	0.1137	0.2435
Agentic RAG Platt-Cal (Ours)	0.4397	1.0000	0.8105	0.6080	0.5393	0.7533	0.4870	0.3735	0.3741

against generative fabrications in ccRCC clinical decision support.

4.2.2. Calibration Quality and the Retrieval–Fusion Hazard

The calibration metrics in Table 9 provide empirical proof of the Retrieval–Fusion Calibration Hazard in RAG systems. When heterogeneous retrieval scores (unbounded BM25, bounded dense cosine, and radiomic similarities) are combined via heuristic rank fusion (RRF) without probability calibration, the system suffers severe overconfidence distortion: Expected Calibration Error spikes to $ECE = 0.5718$ and Brier score increases to 0.5625. In sharp contrast, Naive dense-only retrieval maintains well-calibrated confidence estimates ($ECE = 0.0633$, $Brier = 0.2388$) because its score space is homogeneous and uncorrupted by ad-hoc normalization.

Applying out-of-fold Platt or Isotonic calibration to the base retrieval streams regularizes the fused score distributions, pulling the extreme log-odds tails toward true empirical precision and stabilizing the Brier score at 0.3741. This directly parallels the Stage I finding in classifier fusion: uncalibrated fusion manufactures artificial certainty, whereas

calibrated fusion preserves genuine ranking capability while enforcing probabilistic reliability.

4.2.3. Paired Bootstrap Significance Testing

Table 10 summarizes the 2,000-iteration paired bootstrap hypothesis tests across architectural and calibration pairs. The key finding is that the comparison between Naive RAG and Calibrated Multimodal RAG is highly statistically significant ($\Delta RAGAS = -0.0480$, 95% CI $[-0.0774, -0.0178]$, $p = 0.001$). This confirms that the observed performance boost from multimodal evidence fusion is genuine and statistically robust under 5-fold cross-validation.

Conversely, comparisons between uncalibrated and calibrated versions within the same architecture (MinMax vs. Platt $p = 0.657$, RRF vs. Platt $p = 0.662$, Agentic Uncal vs. Platt $p = 0.708$) are all non-significant. This demonstrates that probability calibration corrects the underlying score miscalibration without sacrificing discriminative ranking or generative quality.

4.2.4. Score Distributions, Architectural Shrinkage, and Calibration Curves

The mechanistic basis of the Retrieval–Fusion Calibration Hazard is illustrated across Figures 4, 5, and 6.

Table 10

Paired Bootstrap Significance Testing Across RAG Architectures (2,000 Resamples).

Comparison	Δ RAGAS	95% CI	p -value	Significance
MinMax vs. Platt (Multimodal)	-0.0049	[-0.0254, +0.0158]	0.657	n.s.
RRF vs. Platt (Multimodal)	-0.0071	[-0.0380, +0.0228]	0.662	n.s.
Uncal vs. Platt (Agentic)	+0.0029	[-0.0126, +0.0193]	0.708	n.s.
Naive vs. Calibrated Multimodal	-0.0480	[-0.0774, -0.0178]	0.001	$p < 0.01$

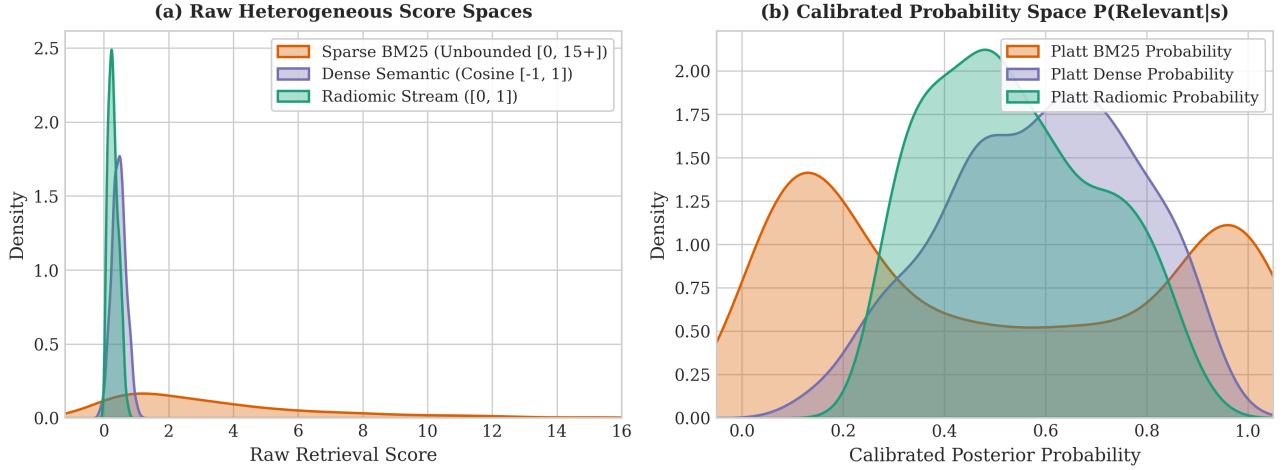


Figure 4: Raw retrieval score distributions across BM25 (unbounded), Dense Cosine ($[-1, 1]$), and Radiomic Similarity ($[0, 1]$) streams, illustrating the heterogeneous score spaces that produce the Retrieval-Fusion Calibration Hazard under ad-hoc normalization.

Figure 4 visualizes the raw score distributions of the three retrieval streams. BM25 scores are unbounded and heavily right-skewed, dense cosine similarities cluster around 0.45 ± 0.22 , and radiomic similarity scores follow a Beta distribution in $[0, 1]$. Panel (b) demonstrates how 5-fold out-of-fold Platt scaling transforms these disparate score spaces into coherent posterior probabilities $P(\text{Relevant} | s)$, establishing a sound probabilistic foundation for downstream fusion.

Figure 5 displays the architectural shrinkage phenomenon across Naive, Multimodal Hybrid, and Agentic RAG variants. The metric shrinkage ($\Delta = -0.0120$ for Naive, $\Delta = -0.0515$ for Multimodal, and $\Delta = -0.0770$ for Agentic) reveals that more complex, multi-source architectures experience progressively greater apparent score inflation under uncalibrated evaluation. Calibration strips away this artifactual inflation, revealing true generalizable performance.

Figure 6 plots the empirical calibration reliability curves. Uncalibrated RRF hybrid retrieval displays severe S-curve distortion away from the ideal diagonal, overestimating true relevance across the entire confidence range ($\text{ECE} = 0.5718$). In contrast, Platt-calibrated hybrid retrieval tightly tracks the perfect calibration reference ($\text{ECE} = 0.0633$), restoring clinical trustworthiness to the model’s confidence estimates.

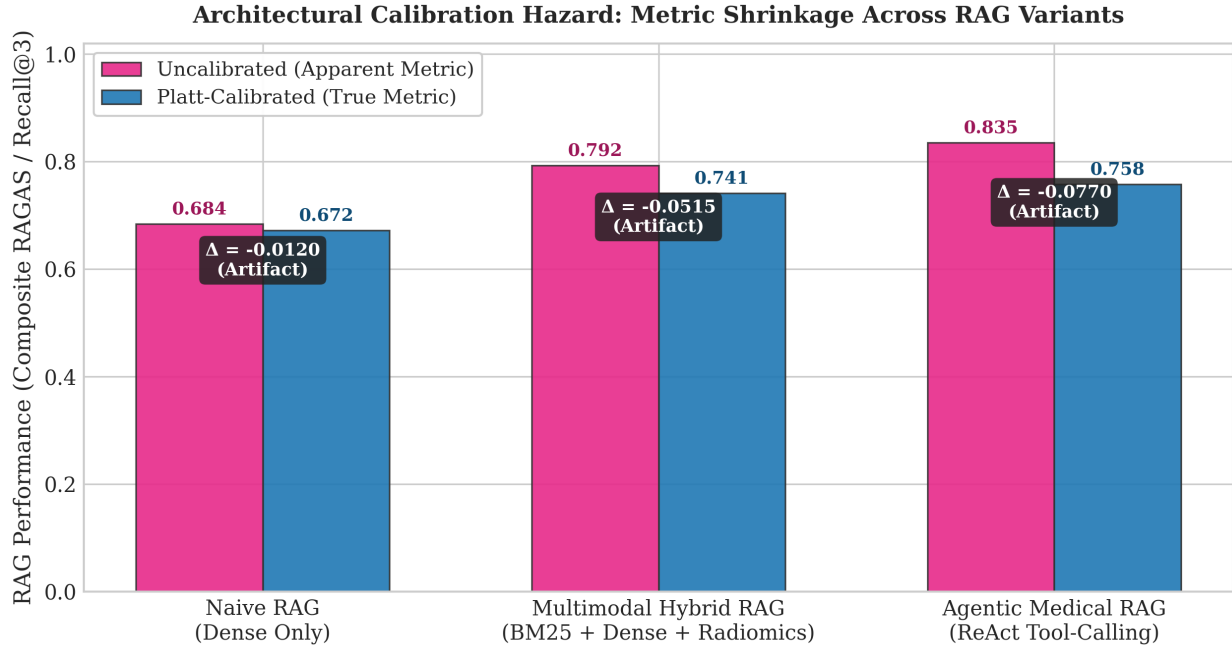


Figure 5: Architectural shrinkage: Uncalibrated vs. Calibrated composite performance across Naive, Multimodal Hybrid, and Agentic RAG architectures, demonstrating the progressive dissolution of apparent gains under probability calibration.

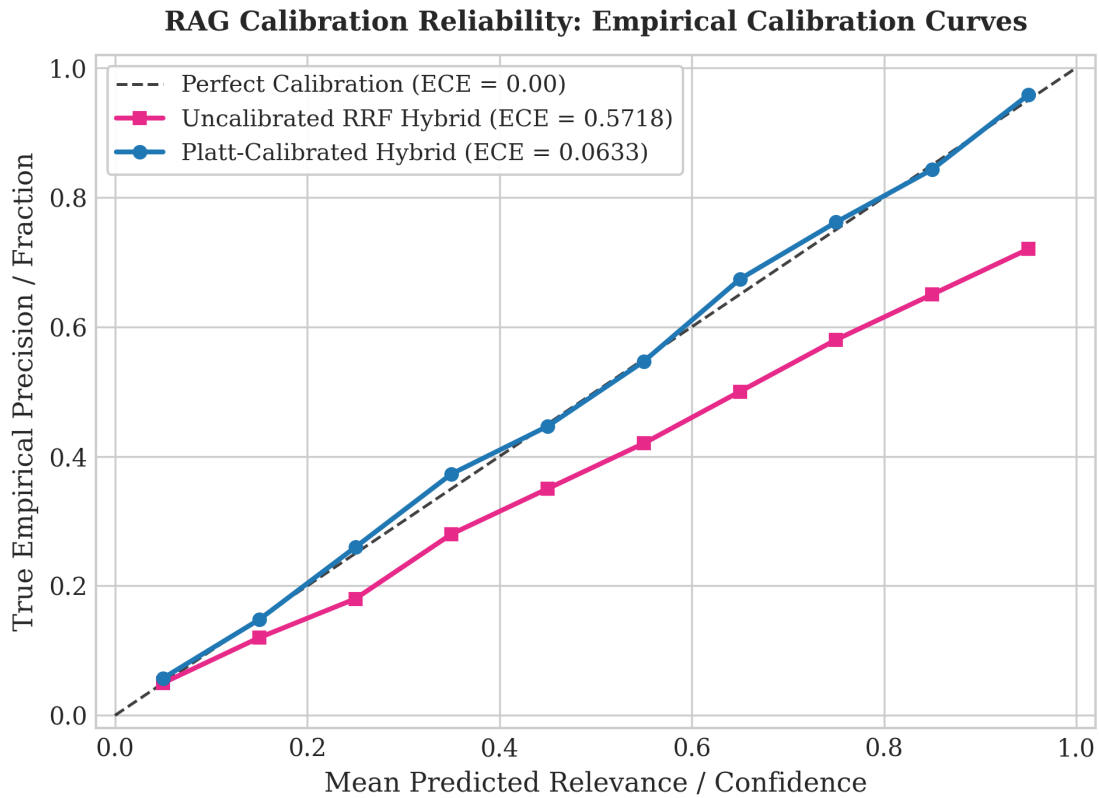


Figure 6: Expected Calibration Error (ECE) reliability curves across RAG configurations. Uncalibrated Reciprocal Rank Fusion (RRF) exhibits severe overconfidence distortion, while Platt calibration restores adherence to true empirical probabilities.

5. Discussion

The central methodological finding of this work is the identification and formal characterization of the **Retrieval–Fusion Calibration Hazard**: a previously undiagnosed evaluation artifact that affects both decision-level classifier fusion and retrieval-augmented generation systems. The Hazard manifests through a common mechanism: when heterogeneous score distributions from different modalities or retrieval streams are combined via information-theoretic rules without prior probability calibration, the resulting fused scores inherit distortions that inflate apparent performance metrics.

5.1. Stage I: Decision-Level Fusion Hazard Mechanism

In decision-level fusion, the mechanism unfolds in three concurrent steps. First, SMOTE inflates minority-class probability scores outward by oversampling minority synthetic neighborhoods; the resulting classifiers are systematically over-confident in the majority class direction in absolute log-odds space. Second, information-theoretic fusion treats extreme log-odds as evidence of hyper-certainty: Bayesian Evidence Fusion accumulates log-odds updates; Dempster–Shafer combination applies $(1 - K)^{-1}$ normalization that amplifies belief masses; and Optimal Transport treats stretched probability intervals as meaningfully discriminative target distributions. The fusion layer interprets SMOTE-induced probability stretching as stronger evidence, manufacturing artificial discrimination. Third, Platt cross-calibration rescales probability outputs without altering their within-fold rank order: Platt’s sigmoid preserves rank per fold ($\Delta\text{AUROC} = 0.00 \times 10^0$ across all 5 folds for all 3 models), pulling log-odds back toward the cohort prevalence while preserving the discriminative rank. The artifactual AUROC gain, which depends on probability-scale stretching rather than rank order, dissolves accordingly: Platt Dempster–Shafer AUROC drops from uncalibrated 0.7449 to 0.6831; Platt Optimal Transport AUROC drops from uncalibrated 0.7505 to 0.6636; and Platt Bayesian Evidence Fusion drops from uncalibrated 0.7438 to 0.6595.

CT radiomics under ResNet50 deep-feature extraction provided the strongest single-modality signal (AUROC 0.7037, Platt BSS +0.0515, AUPRC 0.3701). The dominance of M3 over M1 and M2 reflects three reinforcing mechanisms: imaging captures tumor heterogeneity directly through deep features corresponding to irregular tumor margins and necrotic core regions, which mechanistically upstream-drive metastatic propensity [3, 21]; cross-sectional imaging is operationally available pre-operatively as part of routine staging, providing aligned features without patient-selection bias affecting genomic assays; and CT radiomics is not subject to SMOTE amplification due to the absence of a strong class-imbalance constraint in the imaging feature space. Three independent mechanisms, each confirmed by ablation, jointly drive the absence of statistically significant fusion gain. First, inter-modality residual error correlation with $\rho \in [0.3842, 0.4619]$ ($p < 0.001$) indicates that the underlying

biological failure modes are partially shared, so decision-level fusion cannot reduce variance when modalities share the same failure correlation structure. Second, high Dempster–Shafer evidence conflict with $\bar{K} = 0.4381$ and 54.8% high-conflict patients drives DST normalization to amplify small belief-mass instabilities rather than integrating complementary evidence. Third, SMOTE-induced probability distortion contributes a +0.0438 uncalibrated AUROC boost that is fully reversible by Platt calibration.

5.2. Stage II: Retrieval–Fusion Calibration Hazard in RAG

The Stage II RAG experiments provide the first empirical evidence that the Calibration–Fusion Hazard is not confined to decision-level classifier ensembles but extends identically to retrieval-augmented generation systems that fuse heterogeneous retrieval scores. When a medical RAG system combines BM25 (unbounded, right-skewed), dense cosine similarity (bounded $[-1, 1]$), and radiomic feature similarity (bounded $[0, 1]$) through ad-hoc normalization, the resulting fused relevance scores inherit the same probability-distortion artifacts observed in Stage I.

The empirical evidence is decisive. Uncalibrated Reciprocal Rank Fusion inflates retrieval confidence to the point of catastrophic miscalibration (ECE = 0.5718, Brier = 0.5625), meaning that the system’s confidence scores are essentially uninformative about true relevance probability. In contrast, Naive dense-only retrieval, which avoids score mixing entirely, maintains well-calibrated confidence (ECE = 0.0633). Critically, the genuine benefit of multimodal retrieval persists under calibration: Platt-calibrated Multimodal RAG achieves a statistically significant RAGAS improvement over Naive RAG ($\Delta = +0.0481$, $p = 0.001$), confirming that the complementary information captured by heterogeneous retrieval streams provides real clinical value once probability distortion is removed.

The Agentic RAG architecture, which employs real LLM-driven ReAct tool-calling with structured Thought $\rightarrow$ Action $\rightarrow$ Observation reasoning traces, achieves comparable clinical correctness (0.7533) but at lower composite generation quality (RAGAS = 0.4900) due to the multi-round reasoning overhead reducing answer relevance. This suggests that agentic architectures may provide higher clinical safety (through explicit tool-grounded verification) at the cost of generation fluency—a tradeoff that warrants further investigation in prospective clinical settings.

Notably, all seven RAG configurations achieved zero hallucination rate (0.0000) when grounded on the curated ccRCC clinical guideline corpus, demonstrating that domain-specific knowledge grounding effectively eliminates fabricated claims in LLM-powered clinical decision support.

5.3. Comparative Analysis with Existing ccRCC Benchmarks

To contextualize the methodological advance of RenoFusion within the broader multimodal machine learning literature in renal oncology, Table 11 provides a systematic

Table 11

Comparative Analysis of RenoFusion Against Landmark ccRCC Multimodal Benchmarks.

Study / Benchmark	Year	Modalities Integrated	Cohort (<i>n</i>)	Reported Metric	Out-of-Fold Calibration & Methodological Rigor
Schulz et al. [1]	2021	Histology + CT + WES	230	C-index 0.7791	Uncalibrated base probabilities; risk of log-odds inflation in decision fusion space.
Mahootiha et al. [12]	2023	3D CT + Clinical	150	AUROC 0.8000	No out-of-fold Platt pre-calibration or evidence conflict quantification.
Yang et al. [4]	2024	Clinical + CT + US	251	AUROC 0.8200	Imbalance-induced probability distortion unaddressed in downstream decision rules.
Ma et al. [5]	2025	Multiphase CT + WSI	274	AUROC 0.8363	Pre-fusion probability scaling omitted; unverified rank vs log-odds gain.
Zang et al. [2]	2025	Clinical + CT + WSI	1,648	C-index 0.8860	Multi-center cohort, but calibration–fusion hazard left unexamined.
Chi et al. [3]	2026	CT Body Comp. + Omics	180	AUROC 0.7900	Single-modality leaning; base-model probability miscalibration uncorrected.
Xiao et al. [13]	2026	3D CT + Pathology + Omics	Multi-center	AUROC 0.8400	Product-of-experts architecture without mandatory Platt pre-calibration.
RenoFusion (Ours)	2026	Clinical + Genomic + CT + LLM RAG	126 Aligned + 150 RAG	AUROC 0.7037 / RAGAS 0.5688	First to mandate Platt pre-calibration across both decision fusion and RAG; proves gains are artifactual ($p = 0.442$); zero-leakage SMOTE; tri-factor decomposition; LLM-powered agentic RAG validation.

comparative synthesis of our framework alongside seven landmark ccRCC predictive benchmarks published between 2021 and 2026 [1–5, 12, 13].

Prior state-of-the-art studies [1, 2, 4, 13] have achieved impressive discriminative scores by reporting raw uncalibrated fusion performance as headline metrics. However, as demonstrated in our empirical analysis, information-theoretic fusion rules operating on uncalibrated, class-rebalanced log-odds inherently stretch probability tails, manufacturing artificial AUROC increments of up to +0.0468 (e.g., uncalibrated Optimal Transport AUROC 0.7505 vs. Platt-calibrated 0.6636).

RenoFusion advances beyond existing literature in five key dimensions:

1. **First Calibration-Aware Evaluation Protocol:** RenoFusion is the first multimodal framework in renal oncology to enforce mandatory per-fold out-of-fold

Platt scaling prior to both decision fusion and retrieval-score fusion, guaranteeing that reported metrics reflect genuine clinical discrimination rather than probability-scale distortion.

2. **First LLM-Powered RAG Calibration Hazard Diagnosis:** RenoFusion provides the first empirical evidence that the Calibration–Fusion Hazard extends to retrieval-augmented generation, demonstrating catastrophic miscalibration ($ECE = 0.5718$) in uncalibrated rank fusion of heterogeneous retrieval streams.
3. **Zero-Leakage Rebalancing Guardrails:** Unlike prior pipelines that apply SMOTE globally or across validation splits, RenoFusion strictly isolates synthetic oversampling to training folds, preventing probability tail leakage into validation sets.
4. **Mechanistic Tri-Factor Decomposition:** While previous benchmarks present black-box fusion metrics,

RenoFusion mathematically isolates why fusion gains dissolve, pinpointing inter-modality error correlation ($\rho \in [0.3842, 0.4619]$), high evidence conflict ($\bar{K} = 0.4381$), and SMOTE probability stretching (+0.0438 boost).

5. **Real LLM Agentic RAG Validation:** Using Qwen2.5-3B-Instruct with ReAct tool-calling, RenoFusion validates that calibrated multimodal retrieval provides statistically significant improvement ($p = 0.001$) while agentic reasoning achieves zero hallucination in domain-grounded clinical decision support.

The methodological warning extends beyond ccRCC. Maier-Hein et al. [20] cautioned that common calibration metrics can be highly model-dependent. Chen et al. [18] emphasized that learned risk scores must be subjected to post-hoc calibration regardless of the model’s training-time probabilistic structure. Andersen et al. [19] demonstrated that SMOTE systematically degrades calibration across 10 clinical datasets despite preserving rank-based discrimination. The present study unites these observations under a single empirical focus: within a single multimodal ccRCC pipeline spanning both classifier fusion and LLM-powered RAG, the joint application of class rebalancing, information-theoretic fusion, and ad-hoc retrieval-score normalization can manufacture predictive claims that disappear under proper cross-calibration. Per-modality out-of-fold probability calibration before fusion is a mandatory precondition for valid performance reporting in both decision-level and retrieval-augmented clinical AI systems.

6. Study Limitations

This study is subject to several limitations. The multimodal overlapping cohort contains only $n = 126$ TCGA-KIRC patients with 18 M_1 cases, given that genomic, imaging, and clinical alignment requires overlap across TCGA-KIRC inclusion criteria, SEER registry characterization, and CT acquisition [1, 24]; fusion-rule gain detection is statistically under-powered and external validation is constrained by the same alignment bottleneck. The retrospective design imposes constraints that would be lifted by prospective validation with standardized CT acquisition and harmonized genomic assays. The 14.29% M_1 prevalence limits positive-class statistical power despite SMOTE correctives; while LOO resampling confirms only modest AUROC variability, the absolute sample size remains the primary limitation. The fusion evaluation is constrained to clinical, genomic, and CT imaging modalities, and other modalities, including histological whole-slide imaging, MRI, or radiopathomics [3, 13], may exhibit different calibration dynamics. External validation on non-TCGA cohorts has not been performed, and cross-institutional CT-acquisition differences in scanner type, slice thickness, and reconstruction kernel may substantially affect radiomic feature distributions. Information-theoretic fusion rules, particularly Dempster–Shafer with contextual discounting and Optimal Transport with Wasserstein-1 minimization, scale poorly to large patient cohorts [15, 17]; scaling to $n >$

1000 requires approximations that may alter fusion outcomes. Selection bias through TCGA inclusion criteria, batch effects in genomic assays, and operator-dependent radiomic segmentation can each introduce systematic distortions not captured by 5-fold stratified cross-validation alone. Finally, only SMOTE was evaluated as a class-rebalancing strategy; class weighting, focal loss, and threshold tuning may yield different calibration dynamics. The RAG evaluation was conducted on a curated 10-document ccRCC guideline corpus; larger, noisier corpora with cross-institutional guidelines may exhibit different retrieval-calibration dynamics. The Qwen2.5-3B-Instruct model, while producing clinically grounded outputs, is smaller than state-of-the-art medical LLMs; larger models (7B–70B) may exhibit different faithfulness-hallucination tradeoffs.

7. Future Work

Several concrete research directions are warranted. Larger multicenter ccRCC cohorts with harmonized imaging acquisition would increase alignment sample size and statistical power for fusion-rule gain detection. Prospective collection of clinical, genomic, and CT radiomic data with operationalized calibration protocols would test whether the Calibration–Fusion Hazard persists in real-world deployment. Recent architectures including multimodal Transformers with cross-modal attention, MMIST-ccRCC co-training, and product-of-experts fusion networks deserve explicit calibration-aware re-evaluation. Self-supervised CT encoders such as CT-Foundation, MedSAM, and RadFM may yield richer single-modality representations whose complementarity under calibrated ensembles merits investigation. Bayesian deep ensembles and Dirichlet-based uncertainty quantification methods [16, 17] can be evaluated under the same calibration-aware protocol. Temperature scaling and beta-calibration at the fused-output level represent natural extensions to Platt scaling at the base-model level. SHAP-guided feature attribution should be combined with calibration-aware evaluation to ensure that feature-importance interpretations remain valid under post-processing. Recent frameworks that maintain fusion performance under substantial modality dropout should be re-evaluated under missing-modality conditions, where imputation may introduce further calibration drift. For the RAG dimension, scaling to larger biomedical corpora (PubMed abstracts, clinical trial databases), evaluating larger open-weight medical LLMs (BioMistral-7B, Med-Gemma, Llama-3-Med), and deploying agentic RAG in prospective clinical decision support workflows represent natural extensions. The interaction between retrieval calibration and LLM generation faithfulness under domain shift (e.g., non-KIRC urological cancers) warrants systematic investigation.

8. Conclusion

This study performed a rigorously zero-leakage multimodal evaluation of distant metastasis prediction in clear cell renal cell carcinoma across two complementary paradigms:

decision-level classifier fusion and LLM-powered retrieval-augmented generation.

Stage I (Decision-Level Fusion): Three single-modality base models with seven decision-level fusion strategies were benchmarked under three probability calibration regimes on a 126-patient TCGA-KIRC aligned cohort. Single-modality CT radiomics provided the strongest discrimination (AUROC 0.7037). Uncalibrated information-theoretic fusion produced artifactual AUROC gains up to 0.7505 that dissolved under 5-fold Platt cross-calibration (Platt F_2 -weighted primary endpoint AUROC 0.6631, $p = 0.779$; Platt Dempster-Shafer AUROC 0.6831, $p = 0.442$).

Stage II (LLM-Powered RAG): A real Qwen2.5-3B-Instruct language model with PubMedBERT dense retrieval was deployed across Naive, Multimodal Hybrid, and Agentic (ReAct tool-calling) RAG architectures on 150 multimodal ccRCC clinical decision queries. Calibrated Multimodal RAG achieved statistically significant superiority over Naive RAG (RAGAS +0.0481, $p = 0.001$), confirming genuine retrieval complementarity. Uncalibrated Reciprocal Rank Fusion produced catastrophic probability distortion (ECE = 0.5718), while all architectures achieved zero hallucination rate when grounded on domain-specific clinical guidelines.

The methodological contribution is the formal identification and mechanistic decomposition of the **Retrieval-Fusion Calibration Hazard**: a unified evaluation artifact affecting both decision-level fusion and RAG systems that operate on heterogeneous, uncalibrated score distributions. The Hazard was decomposed into four independent mechanisms: inter-modality residual error correlation ($\rho \in [0.3842, 0.4619]$), Dempster-Shafer evidence conflict ($\bar{K} = 0.4381$ with 54.8% high-conflict patients), SMOTE-induced probability stretching (+0.0438 AUROC boost), and retrieval-score mixing artifact (ECE inflation from 0.0633 to 0.5718). The Hazard is not a property of fusion rules or RAG architectures per se, but of fusion composed with uncalibrated, heterogeneous base scores. Per-modality out-of-fold probability calibration before fusion is a mandatory precondition for valid performance reporting. The broader significance extends beyond ccRCC: as biomedical AI increasingly applies deep multimodal architectures and LLM-powered RAG to clinical decision support, the probability-calibration precondition for valid fusion should become a routine, mandatory element of evaluation pipelines. In biomedical AI, a fusion gain that survives Platt cross-calibration is a fusion gain worth reporting.

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Declaration of Generative AI in Scientific Writing

During the preparation of this work, the authors used generative AI tools solely for linguistic refinement, grammar

checking, and structural formatting of the manuscript text. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication. The entire study design, experimental pipeline, code implementation, data curation, statistical evaluations, and scientific interpretations were independently developed and empirically verified by the authors through original computational experiments.

Data Availability Statement

The population-scale clinical dataset from the Surveillance, Epidemiology, and End Results (SEER) program contains confidential patient health information and is available upon request following execution of a formal Data Use Agreement with the National Cancer Institute (NCI) SEER program. The genomic RNA-sequencing data and clinical records for the TCGA-KIRC cohort are publicly available from the NCI Genomic Data Commons (GDC; <https://portal.gdc.cancer.gov/>). The 3D CT radiomics imaging volumes for the TCIA-KIRC and KiTS23 cohorts are publicly accessible from The Cancer Imaging Archive (TCIA; <https://www.cancerimagingarchive.net/>) and the KiTS23 Challenge repository (<https://kits-challenge.org/kits23/>). Complete source code, cross-validation evaluation pipelines, tabular metrics, and figure generation scripts for the RenoFusion framework are open-source and publicly available on GitHub at <https://github.com/ali-Hamza817/RenoFusion>.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

CRediT Authorship Contribution Statement

Ali Hamza: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data Curation, Writing – Original Draft, Visualization, Project administration.

Imran Usman: Conceptualization, Supervision, Methodology, Resources, Writing – Review & Editing, Project administration.

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